Case study 6: Mirror frame

National 5 · Practical Woodworking · Answers and marking guidance

Award one mark per correct point unless stated. Other valid answers should be credited. Sketches can gain full marks if they show enough detail.

1 (a) Name a tool used to mark out the mitres. (1 mark)

- Mitre square
- Sliding bevel (set to 45°)

Any 1 point.

1 (b) State the angle each part is cut at. (1 mark)

- 45°

1 (c) Describe one way of strengthening the mitre joints. (1 mark)

- Glue and pin / dowels / a spline or veneer key in a saw cut across the corner

2 Name a cramp suitable for holding the mitred frame while the glue sets. (1 mark)

- Mitre cramp
- Band (web) cramp

Any 1 point.

3 Explain why the rebate is cut before the frame is assembled. (2 marks)

- It is cut along the full length of each straight part
- Much easier and more accurate with a rebate plane / router before the corners are joined; after assembly the plane can't reach into the corners

4 (a) Explain why stain alone does not protect the wood. (1 mark)

- Stain only changes the colour; it doesn't form a protective film

4 (b) Describe two stages of preparing the frame before staining. (2 marks)

- Scrape or plane away marks
- Sand with the grain from coarse to fine using a cork block, then remove the dust

5 Describe how a template could help if 50 identical frames were made. (1 mark)

- Every frame marked out the same and quicker, because sizes and angles are only measured once

Total: 10 marks